



Land Use Department

Courthouse Annex

2045 13th Street • 13th & Spruce Streets • Boulder, Colorado 80302 • (303) 441-3930

BOARD OF COUNTY COMMISSIONERS

AGENDA ITEM

November 25, 1997 -- 9:00 AM

Hearing Room, Third Floor, County Courthouse, Boulder

PUBLIC HEARING

STAFF PLANNER: Chris Toebe

STAFF RECOMMENDATION RE:

Docket SU-93-14: DOWE FLATS MINING & RECLAMATION PROJECT: Operations Center Site Plan Amendment

Request: Approval of an amendment of the specific site plan for the temporary / operations center authorized through Resolution #94-81.
Location: At Dowe Flats, east of Lyons, north of U.S. Hwy. 66, west of 53rd Street, in Section 21, T3N, R70W
Zoning: Agricultural (A)
Applicant: Southdown, Inc. c/o John Lohr
Planning Commission: Recommend that project is in compliance -- October 15, 1997

DISCUSSION

This issue has been placed on your agenda for discussion as the year three interim review, to analyze the compliance of the operator with the conditions of approval, review any complaints regarding the operation and examine whether any additional measures are required to insure compliance with the conditions of approval.

Resolution #94-81 was approved and signed by the Board of County Commissioners with 16 conditions of approval. Condition 5 (b) of the Resolution requires interim reviews at the end of years one, three, five, and then every five years thereafter.

The year one review occurred in the Fall of 1995. Resolution #95-239, signed by the Board of County Commissioners on December 14, 1995, found that the applicant is in compliance with the terms, conditions, and commitments of record subject to the impositions of a condition to insure compliance with the cultural resource monitoring requirements. This condition stated that the applicant shall maintain a catalog recording the cultural resource monitoring operation, which shall be available for view during normal business hours by County staff, the Indian Peaks Chapter of the Colorado State

Jana L. Mendez
County Commissioner

Ronald K. Stewart
County Commissioner

Paul Danish
County Commissioner

Historical Society, or the State Archaeologist. The catalog should include the dates on which a monitor was present; an associated map of the area covered by the monitor's survey; and documentation of findings including the location of the findings, and the manner in which the findings were recorded, curated, and reported to the State Archeologist.

Full production at the facility commenced on August 18, 1997. Approximately 6,500 tons of rock per operating day (45,000 tons per month), are being crushed and transported via the conveyor system to the plant on the south side of State Highway 66. The facility operates 4 to 5 days per week (Monday through Thursday or Friday), 8 to 10 hours per day.

Since the year one review, the applicant has:

- Relocated County Road 47 and dedicated the required trail corridor.
- Constructed screening berms along the relocated County Road 47.
- Agreed to work in conjunction with the County Transportation Department to pursue an amendment to the Vacation of County Road 49.
- Dedicated the Indian Mountain property to Boulder County on September 30, 1997.
- Stockpiled overburden and topsoil for future reclamation activities.
- Obtained the necessary Site Plan Review approvals from the Board of County Commissioners for the Temporary Operations Center, garage and fueling area.
- Constructed the crusher at the south end of the first ridge.
- Constructed the conveyor system and tube crossing State Highway 66 per the County approval. Landscaping and berming is currently occurring adjacent to the system.
- Has conducted and submitted to Boulder County the required reports and updates including pre-blast surveys, blast monitoring data including special monitoring for the Montgomery School and the City of Longmont's water storage tank (both of which experienced only very low vibration measurements (.25 inches per second), reinternment of artifacts, hydrologic studies relative to wetland areas, winter raptor studies, monitoring reports for the Montgomery School, Cultural Resources Management Plan Addendum, annual field inspection report of archaeological properties.

During the construction phase of the project, the applicant indicates that dust control consisted of spraying roads and construction areas with water and applying Magnesium Chloride to permanent, unpaved roads. Now that the quarry is in full operation, additional measures include the use of dust collectors at each transfer point on the conveyor system, and the use of an adjustable stacker belt and fogging spray at the discharge point. Per the conditions of approval, loading and hauling will cease if wind speeds exceed 30 miles per hour.

Noise levels have been monitored with the facility at full operation. Per the materials submitted, operations are within the allowed 55 db noise threshold. The applicants were unable to utilize alternative back-up beepers during daylight hours due to the requirements of the Mine Safety and Health

Administration.

REFERRAL RESPONSES

This docket was referred out to agencies, as updated from the original referral list, and adjacent property owners within 1,500 feet of the site. Referral responses are attached for your review.

The County Parks and Open Space Department indicate that they are in receipt of numerous status reports including the reinternment of artifacts, hydrologic studies relative to wetland areas, winter raptor studies, monitoring reports for the Montgomery School, Cultural Resources Management Plan Addendum, annual field inspection report of archaeological properties.

Two responses from adjacent property owners were received. One response indicates concern over dust problems. The second response indicates concern over the maintenance of County Road #47 (maintenance of this road is the responsibility of the County).

The County Transportation Department, Town of Lyons, City of Longmont, Lyons Historical Society, and the Sierra Club responded "No conflicts."

PLANNING COMMISSION

The Planning Commission reviewed this proposal in a public hearing held on October 15, 1997. In the hearing the applicant provided an overview of quarry operations and mitigation measures that have been implemented. The Planning Commission inquired about plant materials that were being utilized, who was performing the archeological monitoring, how the soil was originally stripped and what modifications were made to the dust collectors. The applicant provided answers to these questions by indicating that a dryland pasture mix is used for seeding, Walker Nightwalker is still the archaeological monitor, and that sealing and padding as well as different bag types have been utilized on the dust collectors.

No public spoke regarding this item, and no new information was provided. The Planning Commission commended and complimented Southdown on being good neighbors, and unanimously passed a motion recommending that the project be found in compliance with the Special Use approval.

SUMMARY / CONCLUSION

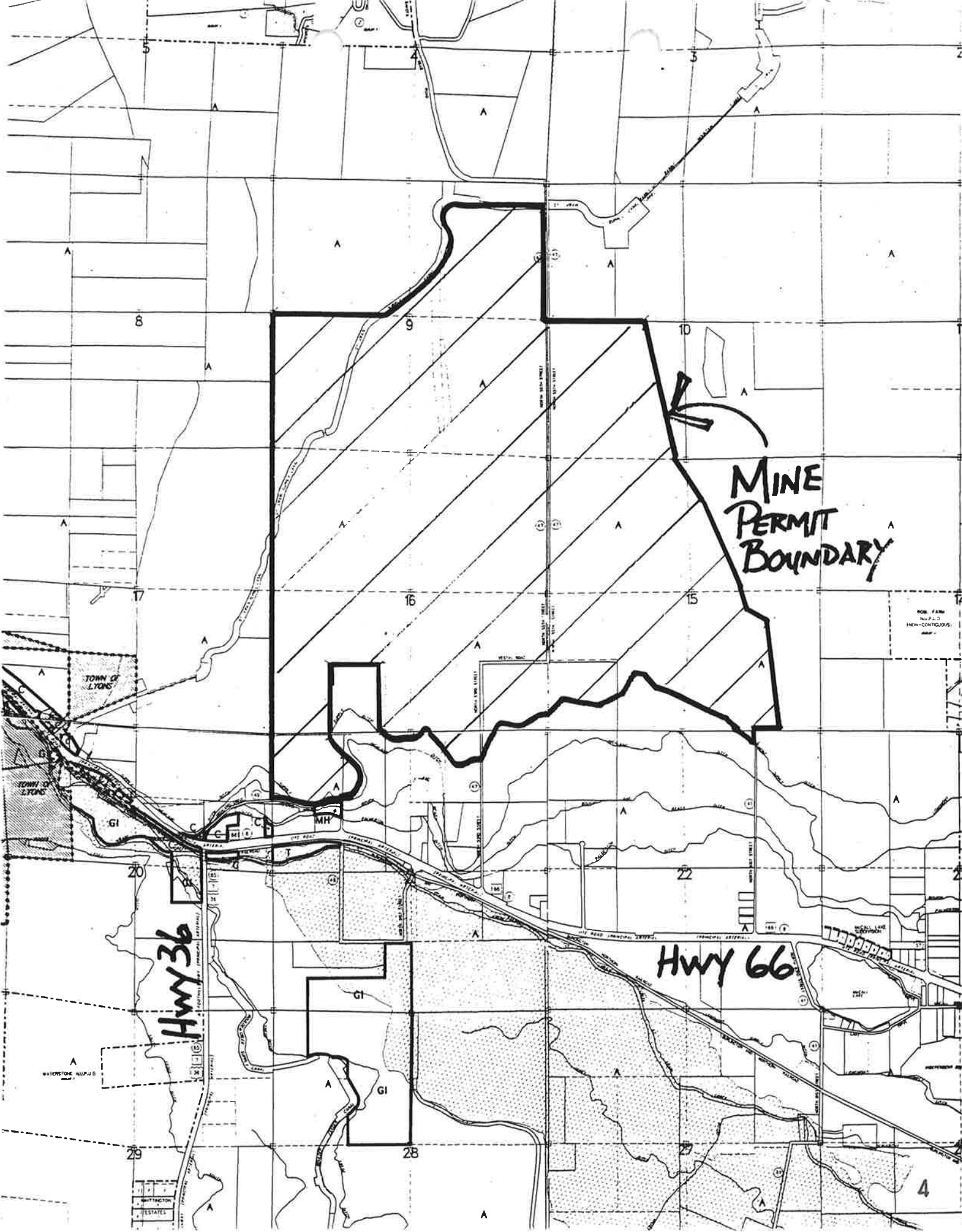
From the review and analysis of the special use file, the applicants submittal, and the referral responses, staff finds that the applicants are operating in compliance with the conditions of approval for Docket SU-93-14. The facility has only recently gone into full production, and the applicant is continuing to work with the County regarding any concerns that arise. The applicant has acquired all applicable permits, and provides all required reports in a timely manner.

RECOMMENDATION

Therefore, the Land Use staff and the Planning Commission recommend that the Board of County Commissioners find that Docket SU-93-14: DOWE FLATS Mining and Reclamation Project continues to

operate IN COMPLIANCE with the conditions of approval, and that the conditions of approval continue to be enforceable.

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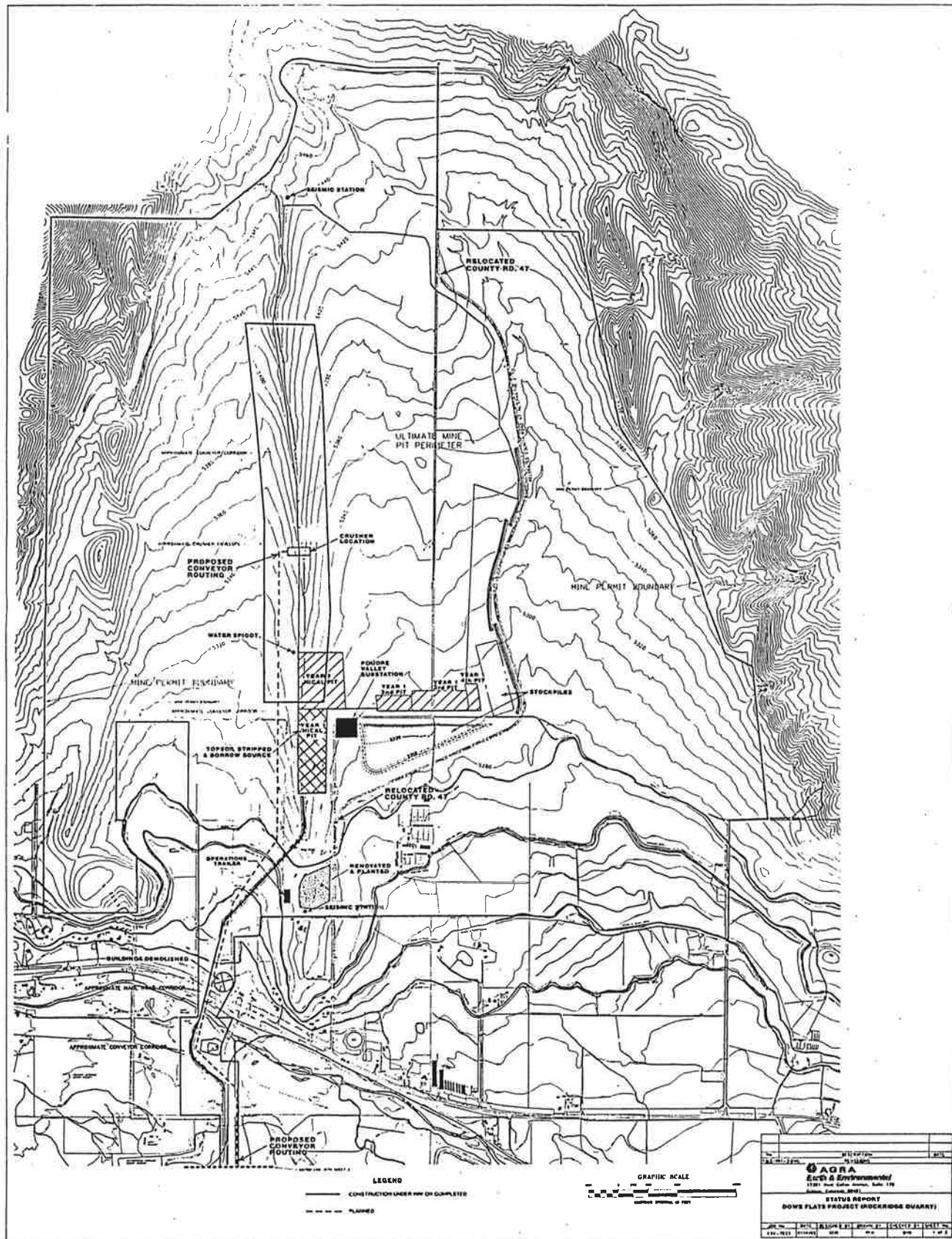


MINE
PERMIT
BOUNDARY

Hwy 36

Hwy 66

4



ATTACHMENT 1

RESOLUTION 95-239

A RESOLUTION FINDING COMPLIANCE (FIRST-YEAR INTERIM REVIEW) WITH BOULDER COUNTY LAND USE DOCKETS #SU-93-14 ("DOWE FLATS MINING & RECLAMATION PROJECT") AND #V-93-8 ("DOWE FLATS VACATION/RELOCATION OF COUNTY ROAD 47 AND VACATION OF A PORTION OF COUNTY ROAD 49"): A REQUEST FOR A SPECIAL USE PERMIT AND ASSOCIATED SITE SPECIFIC DEVELOPMENT PLAN FOR A LIMESTONE/SHALE OPEN MINING/QUARRYING OPERATION ON 312 ACRES WITHIN A 1911-ACRE PERMIT AREA, AND A REQUEST TO VACATE AND RELOCATE THE EXISTING ALIGNMENT OF COUNTY ROAD 47 AND TO VACATE PORTIONS OF COUNTY ROAD 49, ON PROPERTY LOCATED AT DOWE FLATS, EAST OF THE TOWN OF LYONS, NORTH OF U.S. HIGHWAY 66, WEST OF RABBIT MOUNTAIN, IN SECTIONS 9, 10, 15, 16, 21 & 22, T3N, R70W.

WHEREAS, by Resolution 94-81 of the Board of County Commissioners ("the Board") adopted June 28, 1994, Southdown, Inc., a Louisiana Corporation, through its agent John Lohr ("Applicant"), received approval for a special use permit, with associated site specific development plan (Dockets #SU-93-14/V-93-8 ("the Approval")), to mine limestone and shale on a 312-acre area within a 1911-acre permit area on the property which is generally known as "Dowe Flats" and which is located as described in the caption to this Resolution, above ("the Subject Property"), in the Agricultural Zoning District in unincorporated Boulder County; and

WHEREAS, the Approval required that interim reviews of the mining operation occur (as measured from the date of the Board's adoption of Resolution 94-81) at the end of years one, three, and five, and then occur every five years thereafter; and

WHEREAS, the purpose of the interim review is to review the monitoring/management plans and compliance with the terms, conditions, and commitments of record of the Approval; and

WHEREAS, the first-year interim review period under the Approval ended in the summer of 1995, with a public hearing before the Boulder County Planning Commission ("the Planning Commission") on the Applicant's first-year interim compliance being held on October 18, 1995; and

WHEREAS, the Planning Commission found the Applicant to be in compliance during the first-year review period, subject to imposition of a condition to insure adequate cultural resource monitoring pursuant to Condition #14 of the Approval; and

WHEREAS, on November 21, 1995, the Board a duly noticed public hearing on the Applicant's first-year interim compliance, at which hearing the Board considered the recommendation of the County Land Use Department Planning Staff dated November 21, 1995, with its attachments; the testimony of the County Land Use Department Planning Staff; and the documents and testimony presented by the Applicant; the Applicant's environmental studies consultant; and the President of the Boulder County Audubon Society; and

WHEREAS, based on the Public Hearing, the Board finds that the Applicant is in compliance with the terms, conditions, and commitments of record of the Approval, subject to imposition of the condition stated below to insure adequate compliance with the cultural resource monitoring requirements of Condition #14 of the Approval.

NOW, THEREFORE, BE IT RESOLVED that the Applicant is in compliance with the terms, conditions, and commitments of record of the Approval (Resolution 94-81), based on this first-year interim review, subject to imposition of the following condition on the special use permit to insure compliance with the cultural resource monitoring requirements of Condition #14 of the Approval:

The Applicant shall maintain a catalog recording the cultural resource monitoring operations, which shall be available for view during normal business hours by County staff, the Indian Peaks Chapter of the Colorado State Historical Society, or the State Archaeologist. The catalog should include the dates on which a monitor was present; an associated map of the area covered by the monitor's survey; and documentation of findings including the location of discovery, the graded area which was the source of the findings, and the manner in which the findings were recorded, curated, and reported to the State Archaeologist.

BE IT FURTHER RESOLVED that all of the terms, conditions, and commitments of record of the Approval (Docket #SU-93-14 and Resolution 94-81) remain in full force and effect.

A motion to find the Applicant in compliance with the Approval, based upon the first-year interim review, and subject to the condition stated above to insure adequate cultural resource monitoring under Condition #14 of the Approval, was made by Commissioner Mendez, seconded by Commissioner Danish, and passed by a 3-0 vote of the Board.

ADOPTED this 14th day of December, 1995, nunc pro tunc
the 21st day of November, 1995.

BOARD OF COUNTY COMMISSIONERS
OF BOULDER COUNTY:

Ronald K. Stewart
Ronald K. Stewart, Chair

Jana L. Mendez
Jana L. Mendez, Vice Chair

Paul D. Danish
Paul D. Danish, Commissioner

ATTEST:

Juan M. Ashcraft
Clerk to the Board



RESOLUTION 94-81

A RESOLUTION CONDITIONALLY APPROVING BOULDER COUNTY LAND USE DOCKETS #SU-93-14 ("DOWE FLATS MINING & RECLAMATION PROJECT") AND #V-93-8 ("DOWE FLATS VACATION/RELOCATION OF COUNTY ROAD 47 AND VACATION OF A PORTION OF COUNTY ROAD 49"): A REQUEST FOR A SPECIAL USE PERMIT AND ASSOCIATED SITE SPECIFIC DEVELOPMENT PLAN FOR A LIMESTONE/SHALE OPEN MINING/QUARRYING OPERATION ON 312 ACRES WITHIN A 1911-ACRE PERMIT AREA, AND A REQUEST TO VACATE AND RELOCATE THE EXISTING ALIGNMENT OF COUNTY ROAD 47 AND TO VACATE PORTIONS OF COUNTY ROAD 49, ON PROPERTY LOCATED AT DOWE FLATS, EAST OF THE TOWN OF LYONS, NORTH OF U.S. HIGHWAY 66, WEST OF RABBIT MOUNTAIN, IN SECTIONS 9, 10, 15, 16, 21 & 22, T3N, R7OW.

WHEREAS, Southdown, Inc., a Louisiana Corporation, through its agent John Lohr ("Applicant"), has requested approval for a special use permit, with associated site specific development plan, to mine limestone and shale on a 312-acre area within a 1911-acre permit area on the property which is generally known as "Dowe Flats" and which is located as described in the caption to this Resolution, above ("the Subject Property"), in the A-Agricultural Zoning District in unincorporated Boulder County; and

WHEREAS, the Applicant has also requested approval of a vacation and relocation of existing County Road 47, as well as a vacation of portions of County Road 49, which are necessary as part of the mining operation; and

WHEREAS, the mining request includes mining of 100 acres at a time, with extraction of 760,000 tons a year, over a maximum 25-year mining period, with the mining to progress from south to north on the mining site at the rate of 300 feet per year, and also includes the transport of mined material from the mining site to the Applicant's existing cement plant (Southwestern Portland Cement Company) on the south side of U.S. Highway 66, and reclamation of the mining site; and

WHEREAS, the above-described request was processed and reviewed as Boulder County Land Use Docket #SU-93-14/#V-93-8 ("the Docket"), as further described in the Boulder County Land Use Department Planning Staff's Memorandum and written recommendation to the Boulder County Board of County Commissioners ("the Board") dated May 26, 1994, with its attachments ("the Staff Recommendation"); and

WHEREAS, on March 30, 1994, the Boulder County Planning Commission ("the Planning Commission") held a duly-

noticed public hearing on the Docket, and recommended conditional approval of the Docket to the Board, as set forth in the Planning Commission's Certificate of Resolution to the Board dated March 30, 1994; and

WHEREAS, on May 26, 1994, the Board held a duly-noticed public hearing on the Docket ("the Public Hearing"), at which time the Board considered the Staff Recommendation; the recommendation of the Planning Commission; and the documents and testimony presented by the staffs of the County Land Use Department, the County Parks and Open Space Department, and the County Attorney; the Applicant's representatives John Lohr, Paul Banks, and Mike Figgs; and numerous members of the public speaking both for and against the Docket; and

WHEREAS, considering the record of the Public Hearing, the Board finds that, subject to the conditions set forth in this Resolution, below, which are reasonable conditions necessary to allow the Docket to meet the applicable approval criteria, the Docket can be approved, on the basis that the Docket meets the criteria for special use approval and a site specific development plan set forth in Articles 20-301 and 19-301 of the Boulder County Zoning Resolution, and further, the vacation request portion of the Docket will not leave any adjoining land without an established public road or private access easement connecting said land with another established public road; and

WHEREAS, title to the roadways proposed to be vacated in the Docket shall vest in accordance with Part 3 of Article 2 of Title 43, C.R.S., as amended, upon recordation of a resolution of the Board confirming that the Applicant has complied with all applicable terms and conditions for relocating and reconstructing County Road 47 and for vacating County Road 49 as set forth in this Resolution and the Development Agreement to be executed in conformity with this Resolution (the "Development Agreement").

NOW, THEREFORE, BE IT RESOLVED that the Docket is hereby approved, on the basis and terms set forth in this Resolution, above, and subject to the following conditions and commitments:

1. The commitments of record of the Applicant be met, including those listed in the Applicant's summary of May 19/20, 1994, as modified by the following conditions.

2. Non-development Covenants

a. The life of mine non-development covenant shall cover the Permit Property, the West Dowe Flats Donation Parcel, and the Northwest Dowe Flats Property.

b. In the event of a future annexation, the County shall continue to have the right to enforce the non-development covenant, and no waiver or modification of this covenant by an annexing entity shall affect the enforcement rights of the County.

3. Donations

In accordance with the terms and conditions set forth in the Development Agreement, the Applicant will make the following voluntary donations of property to the County:

a. A conservation easement, including appropriate access rights, to protect and preserve the four archaeological sites on the West Dowe Flats Donation Parcel, to be granted contemporaneously with execution of the Development Agreement between the Applicant and the Board.

b. The West Dowe Flats Donation Parcel and the Northwest Dowe Flats Property to be granted for use as open space upon the occurrence of all of the following events: (1) commencement of mining, defined as the movement of at least 100,000 tons of mined material from the permit area to the existing cement plant site; (2) placement of the overland conveyor and crusher into regular operation (operation following start-up testing) in connection with the mining of the permit area; and (3) County acceptance of relocated County Road 47, vacation of old County Road 47, and vacation of portions of County Road 49. The Applicant shall escrow the deeds to the West Dowe Flats Donation Parcel and the Northwest Dowe Flats Property at the time of execution of the Development Agreement. The deeds shall include access rights. A condition of the donation is that no general public access shall be allowed on the West Dowe Flats Donation Parcel in perpetuity, and no general public access shall be allowed on the

Northwest Dowe Flats Property until at least 2030.

c. Contemporaneously with Southdown's dedication to the County of relocated County Road 47, a 25-foot-wide easement on the east side of said relocated road shall be granted to the County for limited uses as a hiking, equestrian, bicycle and non-motorized vehicle trail.

d. Prior to the completion of the conveyance of the donation parcels and the trail corridor to the County, the Applicant shall not convey away any of its rights, title, or interest in easements or other access ways for these properties as specified in the Development Agreement. Upon acquisition of the properties, the County shall not allow general public access to or from them via the Indian Gap Subdivision.

e. In the event of a future annexation, the County shall continue to have the right to enforce the commitment for donation of these properties, and no waiver or modification by an annexing entity shall affect the enforcement rights of the County.

4. Conveyor System

a. A conveyor system shall be used to haul material from the permit area to the existing cement plant. Limited truck haul may be used during the first two years of mining to transport up to 100,000 tons of mined material per year.

b. A new bridge and access road to the existing cement plant will be built, the old bridge removed, and the disturbed land restored. County Land Use Department Staff review and approval of final plans for the conveyor, including color, location, and landscaping within the Highway 66 view corridor, shall occur prior to construction.

c. No more than 200,000 tons of material may be transported to the plant until the conveyor commences operation and the County accepts the construction of relocated County Road 47.

d. In order to mitigate the impacts of the conveyor system, including noise, dust, and vibration, the Applicant shall make a reasonable purchase/option offer to persons owning existing residences within 500 feet of the conveyor system. Such offer shall be based on a reasonable appraisal of the property's value without the Dowe Flats mining operation in place. The offer shall be delivered to the affected property owners or sent to the owners by certified mail, return receipt requested, and shall provide the owners with 60 days (or such longer time as the Applicant may desire) to accept or reject the offer or submit a counteroffer. If the offer is accepted, the Applicant shall purchase the buffer property.

5. Periodic Review

a. The permit shall be subject to both interim and on-going review and assessment. The cost of studies or data produced by the Applicant as part of the required reviews, as well as of the County staff's time to process the reviews, shall be borne by the Applicant.

b. Interim reviews shall occur (as measured from the date of the Board's adoption of this Resolution) at the end of years one, three, and five and then occur every five years thereafter. Interim reviews shall be to review the monitoring/management plans and compliance with the permit terms, conditions, and commitments of record.

c. The Applicant shall also submit to the County Land Use Department copies of the annual reports mandated by the Colorado Mine Land Reclamation Board (CMLRB).

6. If sustained winds exceed 30 MPH at the site, loading and hauling operations will cease until the wind speed drops below 30 MPH. Crushing, conveying, and drilling operations may continue. Wind speed shall be measured at the mine site.

7. County Road 47 is to be relocated to the proposed western alignment through the east valley of Dowe Flats. Relocation of County Road 47 shall require approval of plans by the County Engineer and provision of a Letter of

Credit for 15% of the cost of the relocation to cover a one-year warranty. Upon acceptance by the County of the relocated portion of County Road 47, and upon the Applicant's conveyance of the trail easement described in 3(c) hereof, the County shall vacate the portions of old County Road 47 which have been relocated, said vacation to be confirmed by recordation of a separate Board resolution.

8. Subject to the retention of easements for existing utilities, County Road 49 shall be vacated south from Highway 66 to the Bullock/Southdown property line upon the occurrence of the following conditions: (i) Southdown's construction of a cul-de-sac at the end of and on the public road at said property line, as depicted on Exhibit L to the Development Agreement; and (ii) installation of an emergency access gate at said cul-de-sac. The occurrence of the preceding conditions shall be confirmed by the adoption and recordation of a separate Board resolution vacating such road as soon as possible after the County Transportation Department determines that the foregoing requirements have been met. After vacation is confirmed, Southdown shall allow adjacent property owners emergency access to Highway 66 over the vacated road through its property or over a new road to be constructed thereon.
9. Power line relocation shall be minimal and will coincide with road relocation.
10. The Applicant shall investigate and pursue the use of alternate safety warning devices for operations in the pit to replace the use of MSHA backup beepers.
11. The perimeter of the areas to be disturbed by mining and reclamation operations will be permanently monumented on the ground, with no mining activities to occur outside of this defined area.
12. The Applicant shall obtain all other permits as required and necessary, and shall amend its CMLRB permit to conform to the Board's approval

herein of the proposed "prairie" reclamation plan.

13. Execution of the Development Agreement by the Applicant as approved by the Board in conformity with this Resolution shall be required.
14. Final management plans, including agricultural, prairie dog and cultural resource plans, shall be approved by the County Land Use and Parks Departments' Staffs prior to the commencement of any mining or mining related activities.
15. Once mining has commenced (defined as the movement of at least 100,000 tons of mined material from the permit area to the plant site), the mining operation as approved herein shall be completed in no more than 25 years. The post-mining reclamation phase of the operation shall be completed in three years (or within such longer time only if expressly authorized or required by the Colorado Mined Land Reclamation Board). The maximum time period for all mining-related activity under this permit shall be 31 years.
16. A temporary operations center consisting of a small office/meeting structure, and parking, refueling, and maintenance area, may be located in the permit area only if a specific site plan for the center is reviewed and approved by the Board, and only if the center is removed after mining is completed.

A motion to conditionally approve the Docket, as stated above, was made by Commissioner Page, seconded by Commissioner Hume, and passed by a 3-0 vote of the Board.

ADOPTED this 28th day of June, 1994, in conformity with the Board's discussion on the proposed Resolution at its regularly convened public business meetings on June 9 and 14, 1994, nunc pro tunc the 9th and 14th days of June and the 26th day of May, 1994.

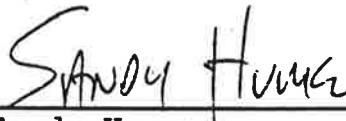
BOARD OF COUNTY COMMISSIONERS OF
BOULDER COUNTY:



Ronald K. Stewart, Chair

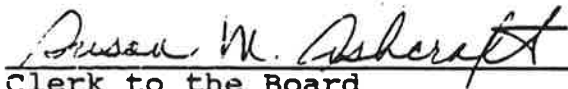


Homer Page, Vice Chair



Sandy Hume

ATTEST:



Clerk to the Board

SOUTHDOWN, INC.

ROCKRIDGE QUARRY AT DOWE FLATS

3 YEAR COMPLIANCE REVIEW

TECHNICAL AND ENGINEERING COMPLIANCE

MANAGEMENT AND MONITORING PLANS

AUGUST 30, 1997
Rev: 9/3/97

INTRODUCTION

The County Commissioners approved the Dowe Flats Mine on June 28, 1994 (Resolution No. 94-81 - Special Use No. SU-93-14 and Vacation No. V-93-8). The Board also approved the Development Agreement for the Project on July 12, 1994. These approvals required a compliance review three (3) years from the date of approval. This document addresses current status and compliance with the key provisions in this approval including mining and reclamation, technical and engineering issues, and management and monitoring three (3) years from the time of approval.

TECHNICAL AND ENGINEERING COMPLIANCE

Mining and Reclamation Plans

Mining operations started in late 1996 and early 1997, and are on-going today. Mine pits have been opened in the 1st, 2nd and 3rd ridges. The 4th ridge pit will be opened shortly. Reclamation was not planned to commence until year three of mining, however overburden and topsoil stockpiles are currently being created for use in subsequent reclamation.

The mine pits and stockpiles are located as shown on the permit maps. Production rock has and will continue to be removed from the stockpile areas so that the resource is removed before it is covered. The mining sequence (generally planned to proceed from south to north) is progressing as permitted except for a temporary change in the mining sequence on the 1st ridge. In order to recover limestone from beneath the crusher location, the planned "year two" mine area is being mined first. After the second year of mining, mining will return to the planned sequence.

Interim Stockpiles

A 5.5 acre overburden stockpile has been created south of the 2nd ridge mine pit. This overburden stockpile will continue to grow over the next 3 years to fill this 25 acre triangle. Topsoil is being stockpiled east the mine area and west of the relocated County Road 47.

Operations Center

The Special Use Permit allowed a mine operations center if Southdown received Site Plan Review approval by the Board of County Commissioners. Two such Site Plan Reviews have been approved. The approved components of the operations center are:

- Office trailer -- Complete
- Fueling area -- Under Construction, Est Comp by 9/30/97
- Garage -- Final Design and Bid, Est Comp by 11/30/97
- Shower room -- Final Design and Bid, Est Comp by 11/30/97
- Parking -- Under Construction, Est Comp by 9/30/97

These land uses are located south and southeast of the mining area in the vicinity of the Supply, Highland, and Rough & Ready ditches.

Crusher

A crusher has been constructed at the south end of the 1st ridge within the envelope shown in the permit application. The crusher is set into the limestone ridge, as planned, and is covered by a metal building to control dust. The slopes around the crusher are currently covered with crushed rock. These will be covered with top soil this fall and planted before Spring. This should help this equipment to blend into the terrain.

Conveyor and Highway 66 Crossing

The conveyor belt and tube over State Highway 66 have been built and are now operational. The conveyor and crossing are located in the corridor shown in the permit application.

The visual impact of the crossing structure was carefully reviewed by the County and architectural approvals were required before construction. Several computer simulations of crossing structure styles were reviewed by the County who selected the style and color that is now present. Landscaping plans for the north and south side of Highway 66, at the crossing, were also submitted to and approved by the County (as well as signage). Landscaping is currently being bid and contracted. Current estimates are that nearly \$70,000 worth of trees and woody plants will be placed in this area. Work should be completed this fall.

A Longmont water line along the south side of Highway 66 was a concern because it was close to one of the planned conveyor support foundations. The City did not want the pipeline damaged and required future access to it. This issue was satisfactorily resolved through discussions with the City on the exact location of the foundation, field inspections by the City during construction, and Southdown granting an easement to Longmont for the water line where an easement did not previously exist.

Maximum Acreage of Disturbance

The Special Use Permit limited the area that could be disturbed by mining and reclamation at any one time to a maximum of 120 acres. Once 120 acres of land has been disturbed, revegetation must commence before additional acreage can be disturbed. Currently, less than 60 acres has been disturbed (as measured from very recent air photos and field inspections). This area will increase over the next 3 years to the maximum acreage allowed. At this point the operation will reach steady state and mining activity will advance northward at a pace of about 300 feet per year.

Berming Requirements

Screening berms were required along the relocated County Road 47 to minimize the views of the mining operation. The initial segment of the berm has been constructed at the south end of County Road 47 and it adequately blocks views of the current mining activity. The berm is currently being extended further north. Further extensions will continue over time as mining advances to the north.

Several additional screening berms are being constructed along the southern boundary of the mine near the new County Road and our neighbors along Highway 66. These berms are an attempt at sight and noise abatement. All should be completed this fall and should be planted before the Spring rain.

Current Operations

The major conveyor construction work was completed in early June. From that time until mid-August, only small volumes of rock were crushed and transported due to numerous electrical and mechanical problems with the system. Full operation commenced during the week of August 18, 1997. Since that time, about 6500 tons of rock has been being crushed and transported each scheduled operating day (a total about 45,000 tons for the month). As of the beginning of September, it appears that we are entering steady-state condition with this system. It is anticipated that we will operate the system for 8 to 10 hours per day, 4 or 5 days per week (Monday-Thursday or Friday).

Road Relocations and Vacations

County Road 47

The Special Use Permit allowed relocation of portions of County Road 47 in Dowe Flats. The relocation and a required trail corridor were dedicated to Boulder County in January, 1995. The abandoned portion of County Road 47 has been vacated by the County and now runs through the center of the active quarry.

3-23-1997 2:20PM FROM SOUTHDOWN LYONS PLAN 3038232199 P. 6

County Road 49

The Special Use Permit required the demolition of the old concrete bridge across St. Vrain Creek on County Road 49, the vacation of County Road 49 up to the Southdown/Bullock property line, and the construction of a "cul de sac" at the Bullock/Southdown property line. In addition, the vacation of County Road 49 was a precondition for donation of the Indian Mountain property to the County Parks and Open Space. This plan was predicated on the continued viability of the old, green, iron bridge which would provide necessary, legal access to adjacent property owners. After the demolition of this old, iron bridge, Southdown worked with the County Public Works Department to develop a new plan to leave the old concrete bridge in place and, at the same time, not request a vacation of County Road 49. This issue is being addressed in a separate process from this third annual review.

Blasting

Blasting commenced at Dowe Flats in March, 1997. Blast warning and schedule signs have been posted on the new County Road 47. All the requirements of the Special Use Permit pertaining to blasting have been complied with.

Pre-Blast Surveys

Southdown has conducted pre-blast surveys on residences and structures in the area to document their condition before the first blast at Dowe Flats. These reports (including photographs) will protect both the neighboring property owners and Southdown in the event of future damage claims.

Blast Monitoring

Southdown has installed two permanent seismic monitoring stations at Dowe Flats, and also has one portable monitor. The permanent stations are located at the northern and southern ends of the mining area. The monitors measure ground motion due to blasting.

To date, ground motion from all blasts has been well under the permit standards.

Special Monitoring - Montgomery School and Longmont Water Tank

Special standards and monitoring provisions were applied to the Montgomery School and the Longmont Water tank. Very low ground vibration standards (i.e. .25 inches per second) were applied to each structure and these have been complied with (the portable seismic monitor has been used at these locations). The "Montgomery School

Monitoring Report, 1997" by Western Historical Studies, Inc. documents this monitoring. Future blasting will move further from these sites, thus further reducing ground vibrations.

Dust Control

Dust control during construction consisted mainly of watering control using the water wagon to spray roads and construction sites. In addition, the permanent, unpaved roads were treated with Magnesium Chloride to help retain moisture and minimize dust generation due to traffic.

As we enter the production phase of the project, several additional measures are required. Dust is typically generated when the material is crushed or dropped. No dusting typically occurs as crushed rock rides the conveyor. First, there are dust collectors on each transfer point on the crusher-conveyor system. Dust collectors function in a very similar manner to a household vacuum cleaner with a fan and a permeable bag. Several of these dust collectors have already undergone major revisions to make them more effective. Changes such as additional skirting and sealing has occurred to make the vacuum fan more effective in pulling in the fugitive dust particles. These transfer points are now controlling fugitive dust well.

another potential fugitive dust generation point is where the rock is discharged from the stacker belt in to the material stockpile. This unit has been designed to raise and lower across a 50 height to control the drop to between 18 and 24 inches. In addition, a fogging spray has been added at the stacker discharge. This combination appears to be very effective even in dry conditions.

The crusher hopper received a special dust control measure, the large metal building covering its entrance. This building allows a haul truck to back into the hopper to dump its 100 ton payload while containing the dust in the building. It is working very effectively.

Wind Speed Monitoring and High Wind Closure

A recording anemometer is located at the mine site and a read out of wind speed is available in the control room at the cement plant. This anemometer has an alarm tied to it to alert the operator whenever sustained winds exceed 30 mph. When this level is attained, loading and hauling will cease (drilling and conveying can continue). No high wind closures have occurred to date.

Noise

Residential noise standards of 55 decibels are required at the Special Use Permit boundary. Engineering Dynamics Inc. measured noise at the mine

under full operating conditions in July 1997. Noise levels from the operation are well below the required standards.

The Special Use Permit also required Southdown to seek alternatives to back up beepers. The Mine Safety and Health Administration was approached on this issue but their regulations absolutely require the use of audible warning beepers during daylight hours (lights can be used at night). Southdown has adjusted the beepers to be as quiet as possible and still be audible above the background noise of the mining operation.

Closure of Larimer County Quarry

Southdown committed to stop shipments of limestone from its Larimer County Quarry after the startup of Dowe Flats. The final loads of rock were delivered to the plant in January, 1997. Since that time, the Larimer County Quarry has been closed and reclaimed.

Other Permit Requirements

Several State and local permits were required to mine at Dowe Flats. All of these have been applied for and/or obtained. They are:

- Colorado Mined Land Reclamation Board - State Permit for mining and reclamation
- Colorado Department of Transportation - permit for access, crossing, and conveyor tube over State Highway 66
- Colorado Department of Public Health and Environment - air emissions permit, water discharge permit
- Colorado Division of Water Resources - Temporary substitute water supply plan
- Boulder County - Floodplain Development Permit

Status and Plans for the Old Plant Quarry

Although there were no permit requirements regarding the old quarry, it is understood that our neighbors have a natural curiosity to know Southdown's plans for this area. A transition from mining to reclamation is taking place in the existing limestone quarries. A small amount of limestone production will continue at the existing quarries but most of the activity will be reclamation. There have been four (4) pits in operation over the last several years: A, B, C, E Pits.

A-Pit at the southeast corner of the property has been filled as a lake covering about 50 acres, as allowed by a County Special Use Permit and a CMLRB permit. Both were approved in the 1980's.. The surrounding shore areas will be graded and planted over the next 18 months.

B-Pit (at the southwest corner near Hygiene Road) has been a Cement Kiln Dust (CKD) disposal pit for the last 10 years. It will be closed and reclaimed within the next 18 - 24 months.

C-Pit at the center of the old quarry will remain as an active mine pit and will be used for CKD disposal over the next 25 years.

E Pit, south of Hygiene Road, has been completely backfilled with overburden and topsoiled. It will be seeded with winter wheat and dryland pasture grasses this fall. This pit is the last location south of Hygiene Road to be mined and reclaimed by Southdown. This property will be sold to Mr. James Guercio this fall and will become part of his Caribou Ranch.

MANAGEMENT AND MONITORING PLANS

The completed Dowe Flats Project Management and Monitoring Plans were submitted to Boulder County on September 12, 1994. The following discussion reviews each of the domains from this document and incorporates other relevant events and information related to the environmental studies for this project that have taken place since the year one review.

Wildlife

An interim report, summarizing field work completed during the fourth year of the five year winter raptor study, has been submitted to Boulder County. Highlights of the report include:

- Field work conducted this past winter sampled transitional activities in both study areas. At the existing Lyons quarries, mining was being completed and large scale reclamation activities were underway. In Dowe Flats, quarry, conveyor and crusher construction was taking place during the entire study period.
- The prairie dog and lagomorph (cottontails and jack rabbits) population and distribution continues to gradually increase following the plague epizootic of 1994.
- Large raptor use in both study areas continues to track the prairie dog prey base.

Data associated with the effects of mining and other human disturbances on winter raptors continues to be collected. The field work for the entire study will be completed in April of 1998. A comprehensive statistical analysis will follow, and a final report will be submitted later in 1998.

Dowe Flats American Indian Advisory Council

The last meeting was held in November of 1995. No meeting was held in 1996 since no major project construction had taken place at that time. The next meeting is scheduled for October 31 and November 1 of this year. The agenda will include a field tour of the mining site, discussion of project construction, reports on the field monitoring of topsoil in the mining area and archaeological sites outside of the mining area, and management of the Indian Mountain property.

Archaeology

Monitoring of topsoil removal in the mining area for prehistoric resources is a requirement of the project Cultural Resources Management Plan. The project field monitor is Walker Nightwalker. He was selected by the plant manager and approved by the Dowe Flats American Indian Advisory Council in 1995. Michael Burney is the backup monitor. Both field monitors have served in these roles since the beginning of the project. Monitoring logs are filed with the plant manager.

Annual monitoring of archaeological sites outside of the mining area on Southdown's buffer properties was conducted in 1996 and 1997 by Marcus Grant, the project archaeologist. The 1996 monitoring report was submitted to Boulder County. The 1997 report is currently being completed.

Historic

As part of a county-wide recordation of historic structures, the Harroun Farm/Garner Dairy was inventoried by cultural resource specialists. Portions of the site were recommended as eligible for nomination to the National Register of Historic Places (NRHP). The Dowe Flats project is anticipated to have minimal impact on this site since the mine operations center was moved from this site to a location approximately one-half mile to the west.

The Montgomery School, an off-site NRHP listed resource, has been monitored by the project historian. The monitoring for 1997 found no change in the condition of the site. Seismographic monitoring during blasting in the mining area found no impacts (zero readings) from blasting. A 1997 monitoring report has been submitted to Boulder County.

3-23-1997 12:23PM FROM SOUTHDOWN LTD'S PLAN 3038232199 P.11

The project historian is also responsible for periodic updates of the project Cultural Resources Management Plan (CRMP). An updated addendum has been submitted to Boulder County. This addendum contains additional details on the archaeology, the Dowe Flats American Indian Advisory Council, and historic domains.

Wetlands

Wetland boundaries were field delineated and marked prior to project construction in Dowe Flats. These boundaries were field checked after mining construction and no impacts to wetlands were documented.

Subsequent to the relining of portions of the St. Vrain Supply Canal in the northwest section of Dowe Flats, additional field analysis was completed on selected wetlands. The Army Corps of Engineers (ACOE) concurs with the project consultant that hydrology has been lost on three wetlands. Other wetlands previously recorded continue to maintain their original status. Copies of the wetlands work and ACOE ruling are on file with the plant manager and Boulder County.

Land Use

Southdown is currently in the process of conveying the "West Dowe Flats Donation Parcel" and the "Northwest Dowe Flats Property" to Boulder County. It is anticipated that this conveyance will be completed by September 30, 1997.

The mine operations center has been moved from the Harroun Farm site (Garner Dairy) to an area adjacent to the southwest corner of the mine quarries. This change was approved by the Board of County Commissioners earlier this year.

Agriculture

No changes in the agricultural operations have been made.

August 13, 1997

Mr. John Lohr
Southwestern Portland Cement
PO Box 529
Lyons, Colorado 80540

Re: Noise Measurements
Southwestern Portland Cement Dowe Flats Limestone Quarry

On Tuesday, July 22, 1997, noise measurements were taken at locations at the Southwestern Portland Cement Dowe Flats Limestone Quarry to verify compliance with the requirements as set forth in Boulder County Special Use Permit 93-14.

This document stipulates that the noise levels associated with mining operations shall not exceed 55 dB(A) at the Mine Permit Boundary.

Noise was measured at the following locations:

1. Site 1
Northeast corner of mine permit boundary near the Rabbit Mountain Trailhead, 24 hour interval.
2. Site 2
At the mine permit boundary directly east of the crusher, 24 hour interval.
3. Site 3
At the mine permit boundary directly west of the crusher, 24 hour interval.
4. Site 4
Southern mine permit boundary adjacent to the meteorological /seismic monitoring station, 24 hour interval.
5. Site 5
At the mine permit boundary coincident with the western property line of the McCain property and directly opposite the conveyor at a distance of 210 feet, 2 hour interval.
6. Site 6
At the mine permit boundary 100 feet west of the meteorological/seismic monitoring station, 2 hour interval.
7. Site 7
At the intersection of the Highland Ditch and the southern mine permit boundary about 480 feet east of County Road, 2 hour interval.
8. At the mine permit boundary, 212 feet in a southeasterly direction, from the 1st transfer station, north of Colorado 66 and at the McCain property fence line, 2 hour interval.

Noise levels associated with the mining operation on Tuesday, July 22, 1997, at these 8 sites, were as follows:

1.	Site 1	Variable and less than 55 dB(A)
2.	Site 2	Variable and less than 55 dB(A)
3.	Site 3	Variable and less than 55 dB(A)
4.	Site 4	Variable and less than 55 dB(A)
5.	Site 5	48 dB(A)
6.	Site 6	46 dB(A)
7.	Site 7	48 dB(A)
8.	Site 8	48 dB(A)

On the day of the sound level measurements, the quarry was running its full fleet on Dowe Flats.

- 4 each, 777 Cat Haul Trucks
- 2 each, 992 Cat Wheel Loaders
- 1 each, D-8 Cat Dozer
- 1 each, Cat Scraper
- 1 each, Rock Drill
- 1 each, Water Wagon

The mobile equipment was conducting overburden removal work in and around the new mine pits. The only mobile equipment currently owned by the Company and not operating on the day in question was the road grader and the explosives truck. In addition to the mobile equipment, the crushing and conveying system was also operating. This is very typical of the equipment and the type of work that will be conducted in Dowe Flats on an "average" day.

The data shows that the noise emissions associated with mining operations at the Southwestern Portland Cement Dowe Flats Limestone Quarry are less than 55 dB(A) and therefore are in compliance with the noise limits defined in the Special Use Permit 93-14.

Prepared By:



Howard N. McGregor
Registered Professional Engineer
State of Colorado
No. 3928

RECEIVED

OCT 15 1997

LAND USE DEPT.

Southdown's Dowe Flats Quarry

3rd Year Compliance Review

Southdown, Inc.

10/8/97

1

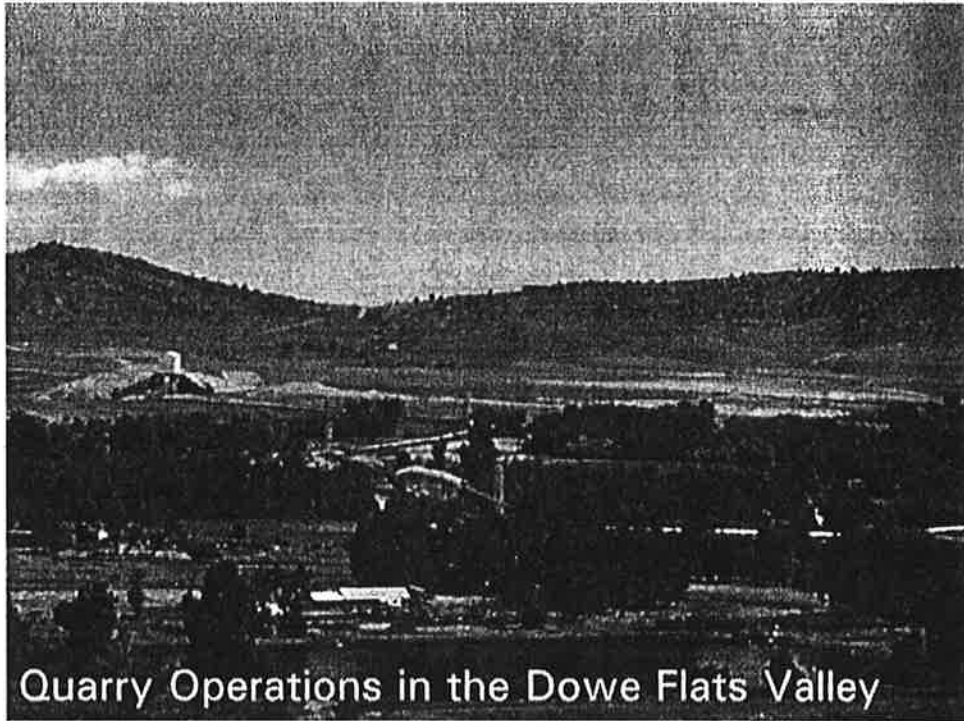
Objectives

- Overview of :
 - ◆ Construction Status
 - ◆ Quarry Equipment & Operations
 - ◆ Current Compliance Status
- Special Use Terms and Conditions
- Completion Plans

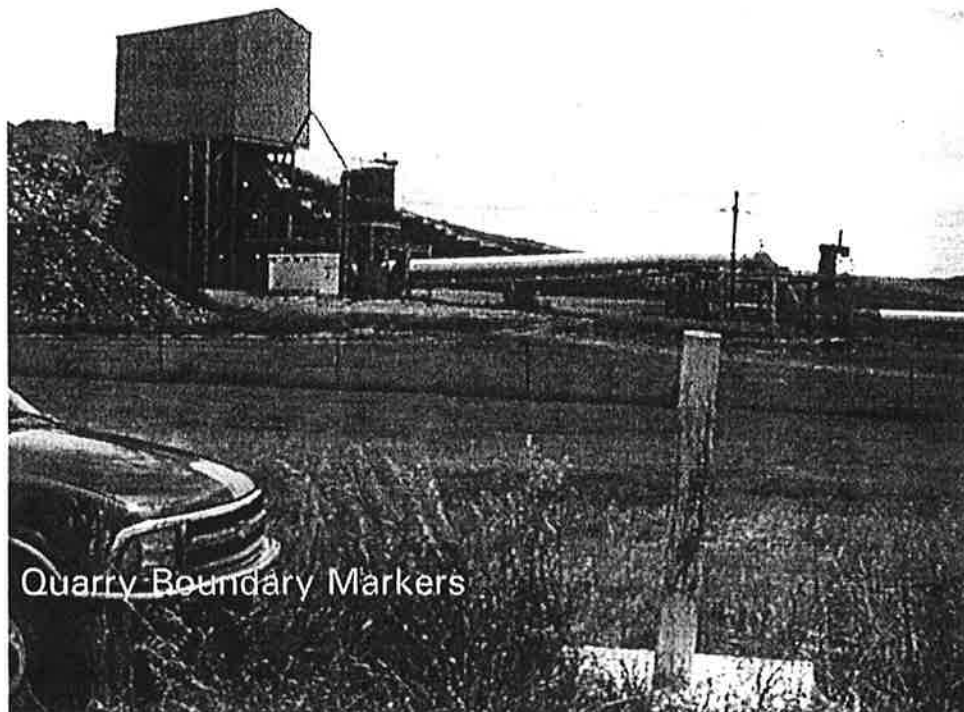
Southdown, Inc.

10/8/97

2



Quarry Operations in the Dowe Flats Valley



Quarry Boundary Markers

Mine Operations

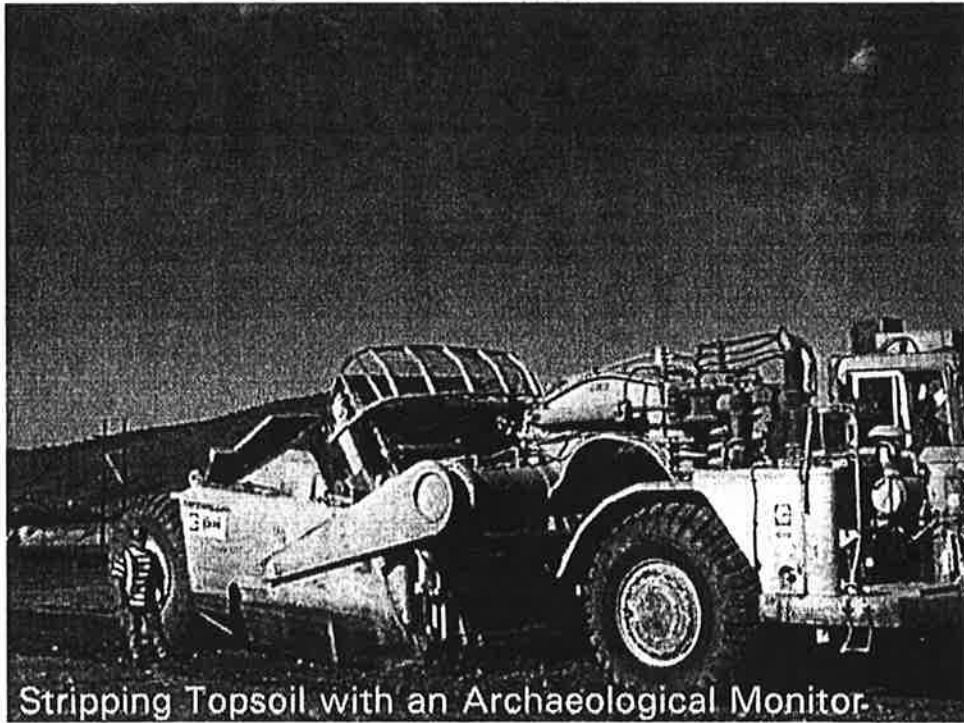
- Soil Stripping with Monitoring
- Drill and Blast
- Remove Overburden & Production Rock
 - ◆ 4 Raw Materials
 - ◆ Wheel Loader & Haul Trucks
- Crush from < 3 feet to < 4 inch
- Conveying at 800 Tons per Hour
- Stacking in Segregated Piles

Southdown, Inc.

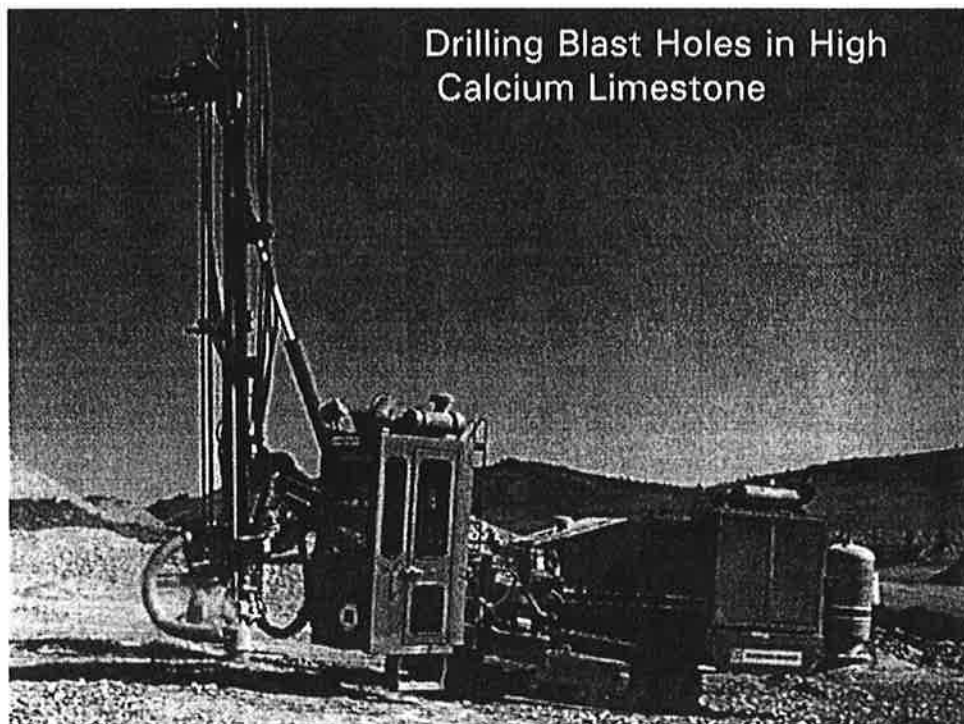
10/8/97

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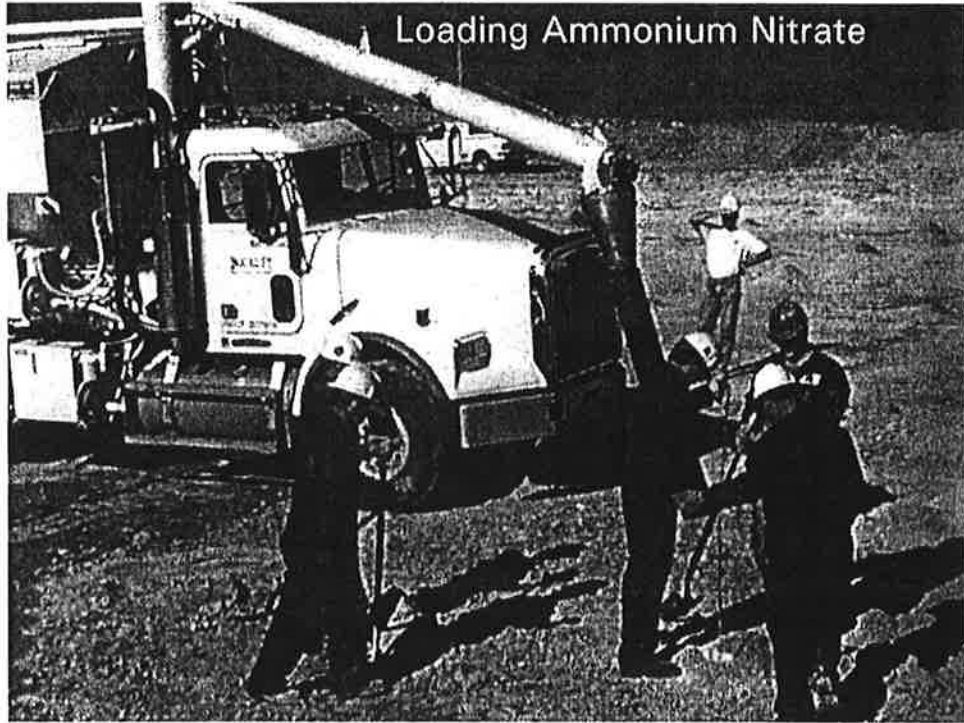




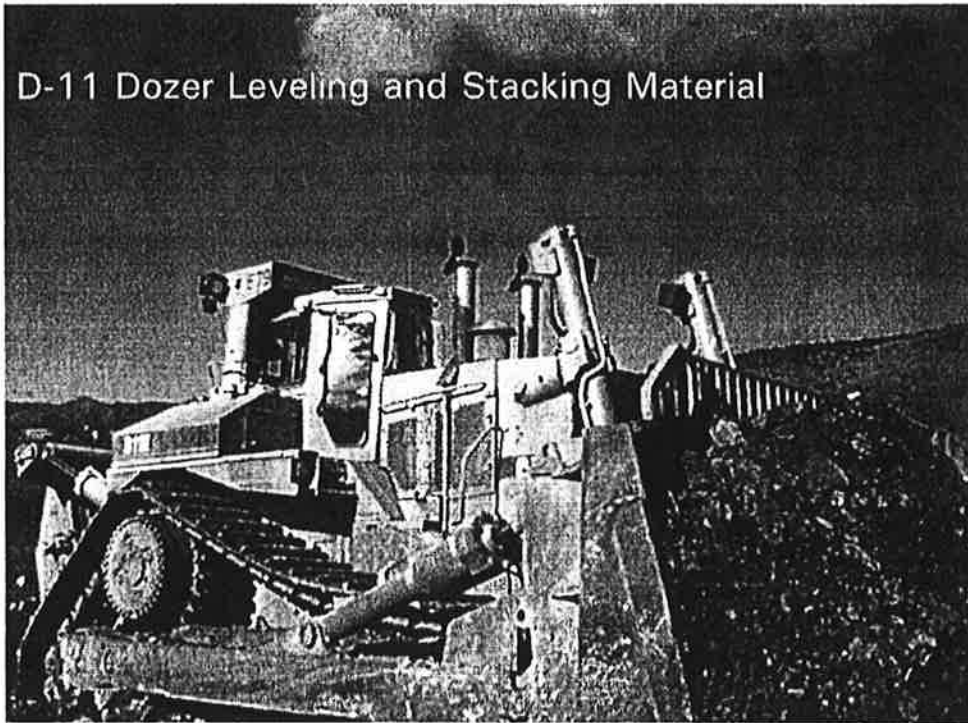
Stripping Topsoil with an Archaeological Monitor.



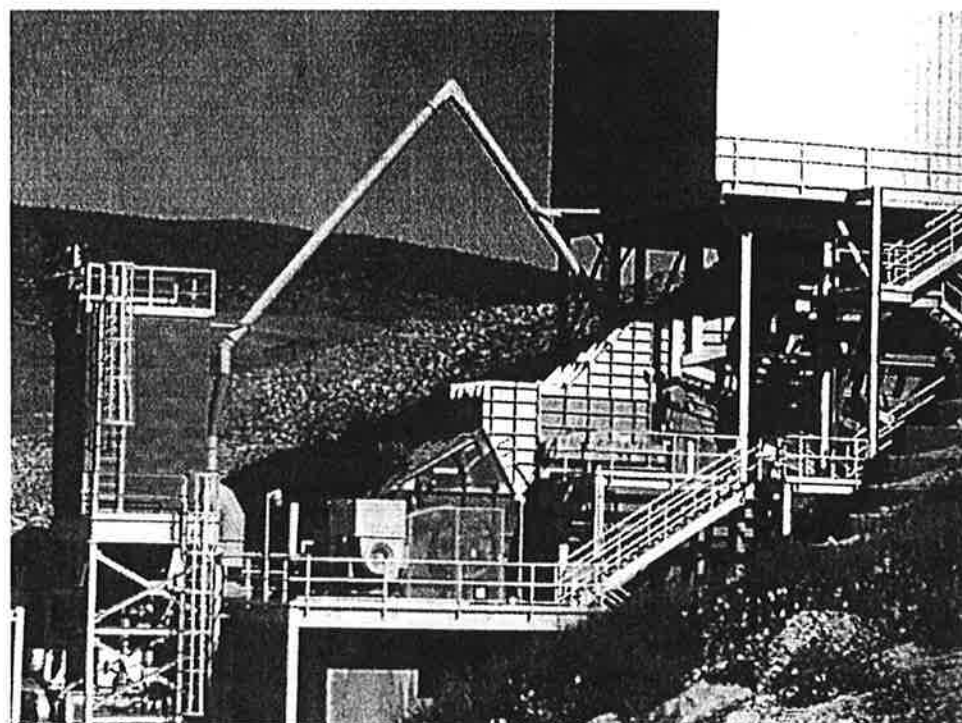
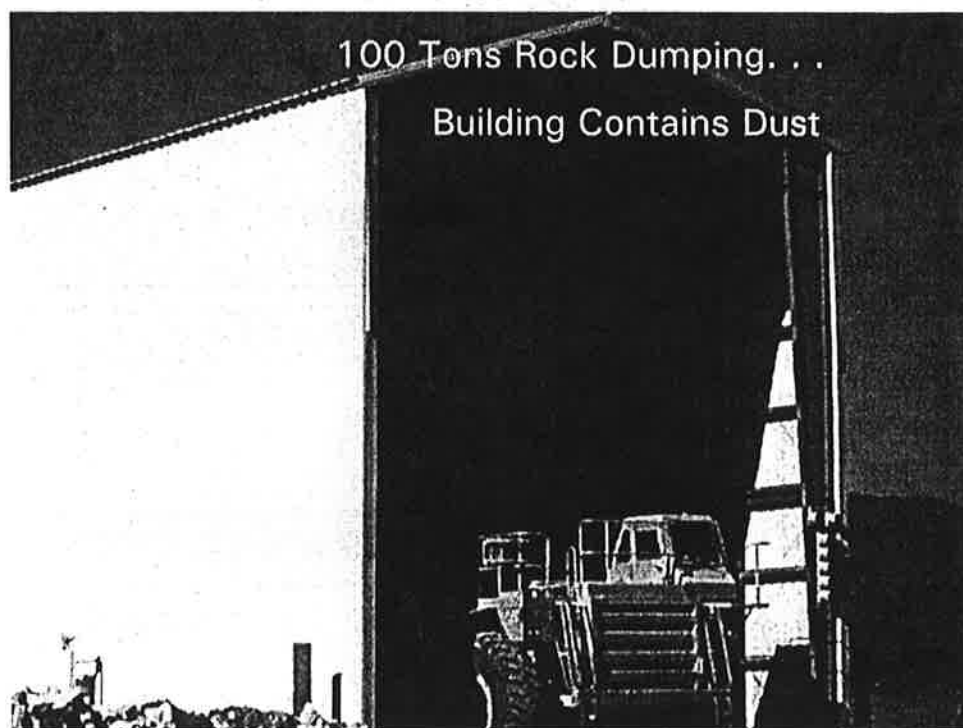
Drilling Blast Holes in High Calcium Limestone

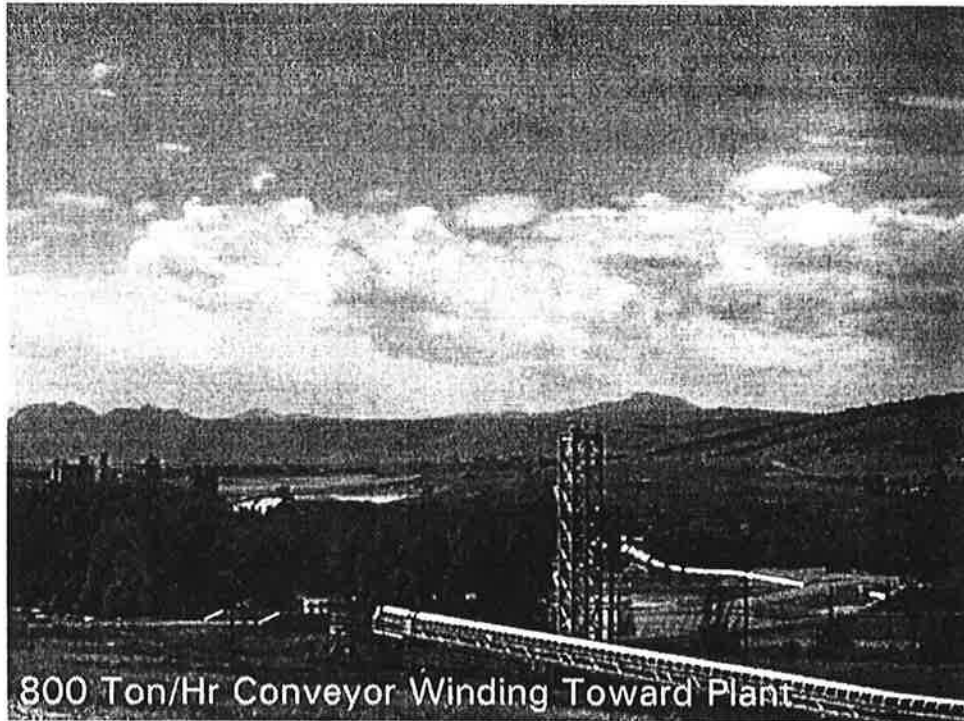


D-11 Dozer Leveling and Stacking Material

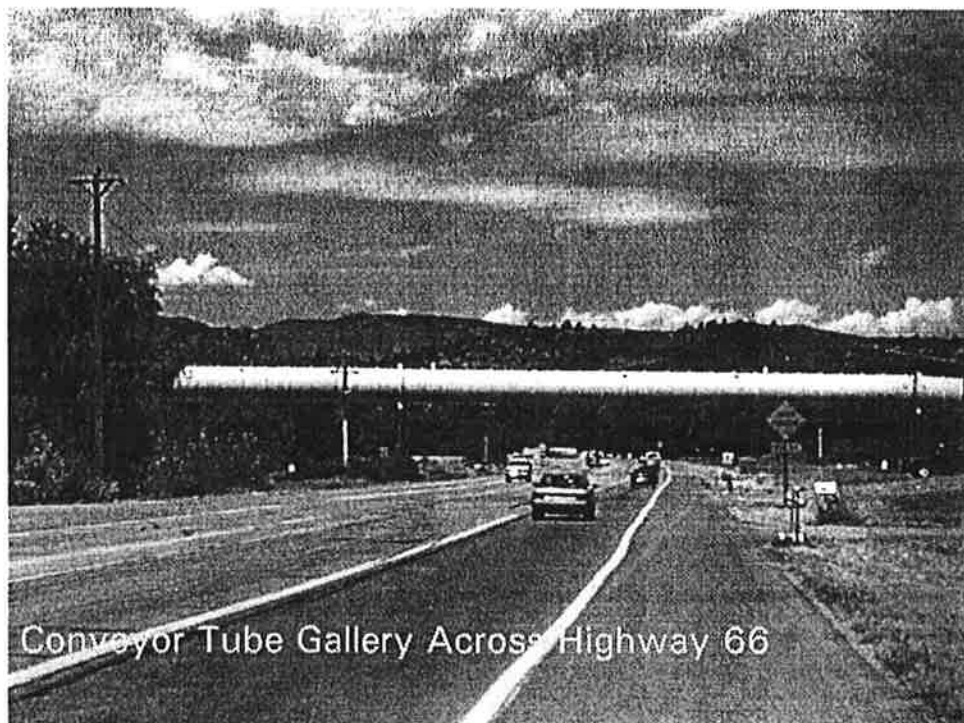


Cat 992 Loading a 777 Cat Haul Truck with 100 Tons of Rock

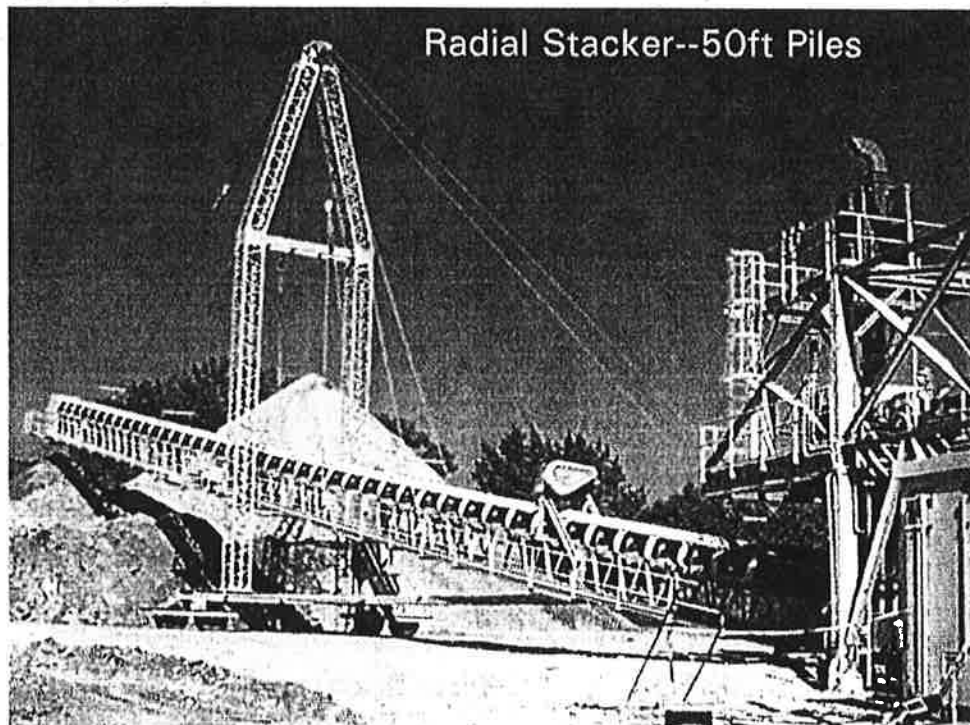




800 Ton/Hr Conveyor Winding Toward Plant



Conveyor Tube Gallery Across Highway 66



Dust Control - Requirements

- No Visible Transport across Boundaries
- Dust Collection on all Transfer Points
- Covers on Conveyors
- Maximum of 120 Acres Open
- Cease Loading and Hauling During Sustained Winds > 30 mph

Dust Compliance

■ Road Dust Control

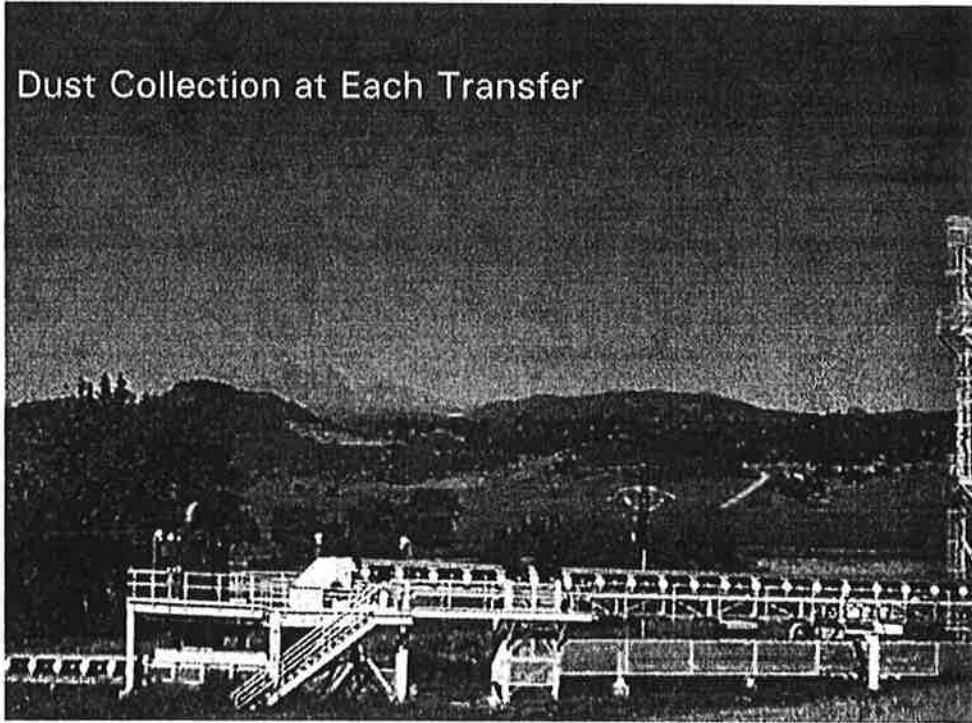
- ◆ Water Wagon when Dry & Above 32 F
- ◆ Mag Chloride On Permanent Roads
- ◆ Water Source at Dowe Flats

■ Dust Collectors at Transfers

- ◆ Ineffective in Dry Conditions
- ◆ Modified or Replaced in Fall, 1997

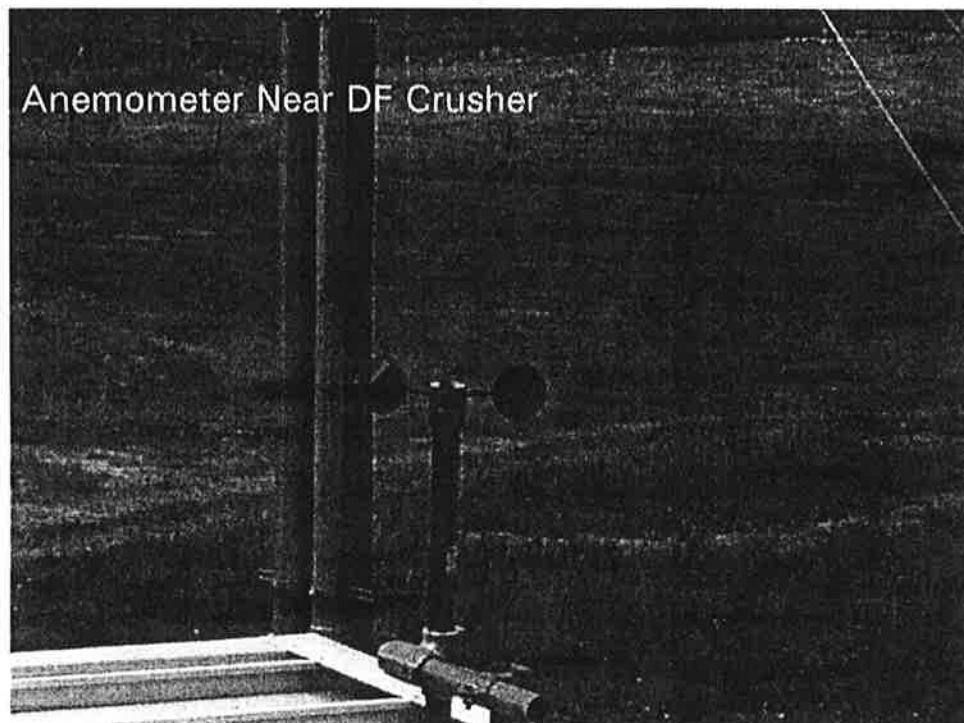
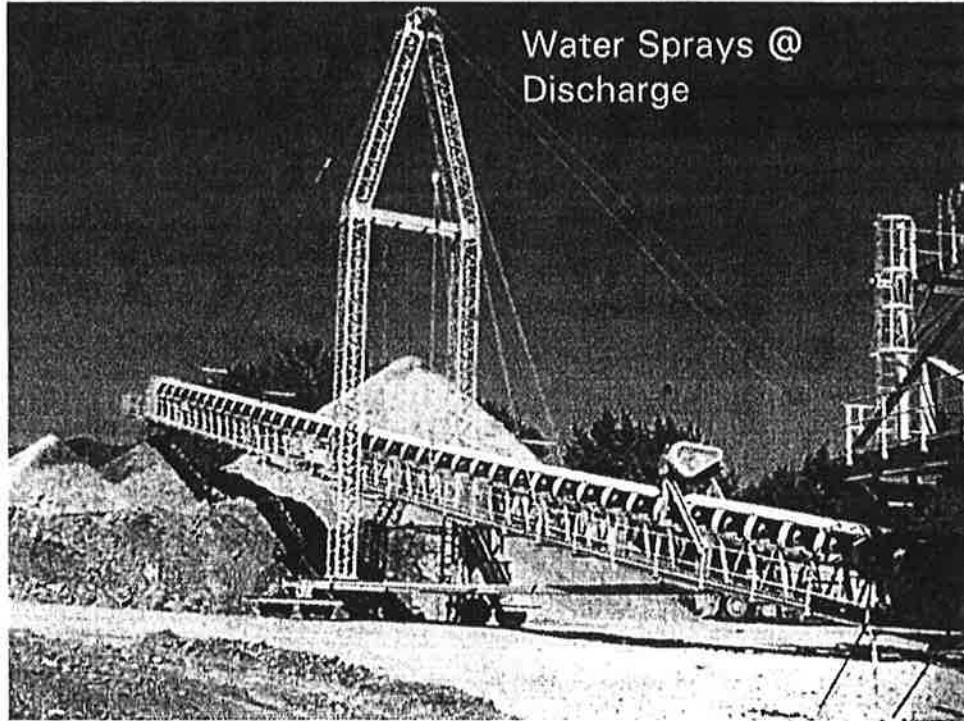


Dust Collection at Each Transfer



Dust Control

- Water Spray and Drop Height on Stacker
- Wind Speed Indicator
 - ◆ Read in Control Room
 - ◆ Set to Alarm ≥ 30 mph for ≥ 15 min
 - ◆ Stop Haulage Activities



Noise Control

- Engineering Dynamics Study (July '97)
- Maximum of 55 dB at Property Boundary
- Maximum Readings = <55 dB
- Typical Readings = 46 - 48 dB
- Ambient Noise Readings > 55 dB
 - ◆ Aerobatic Planes
 - ◆ Highway 66 Traffic
- Noise Berms Completed in Fall of 1997

Southdown, Inc.

10/8/97

25



Noise Control

- Alternatives to Backup Beepers
- MSHA Requires Audible Alarms During Daylight Hours
 - ◆ No Variances due to National Increase in Power Haulage Fatalities
 - ◆ Beepers Tuned Down to a Max of 85 dB from 105 dB

County Road 47

- Road Construction
 - ◆ Designed and Built to County Specifications
 - ◆ Construction Completed in November, 1995
 - ◆ Adopted by County in January, 1996
 - ◆ Letter of Credit Warranty
 - ◆ Trail Corridor Dedicated
- Old County Road 47 Vacated by County
 - ◆ January, 1996
 - ◆ Now Part of Mine Site



County Road 49 Vacation

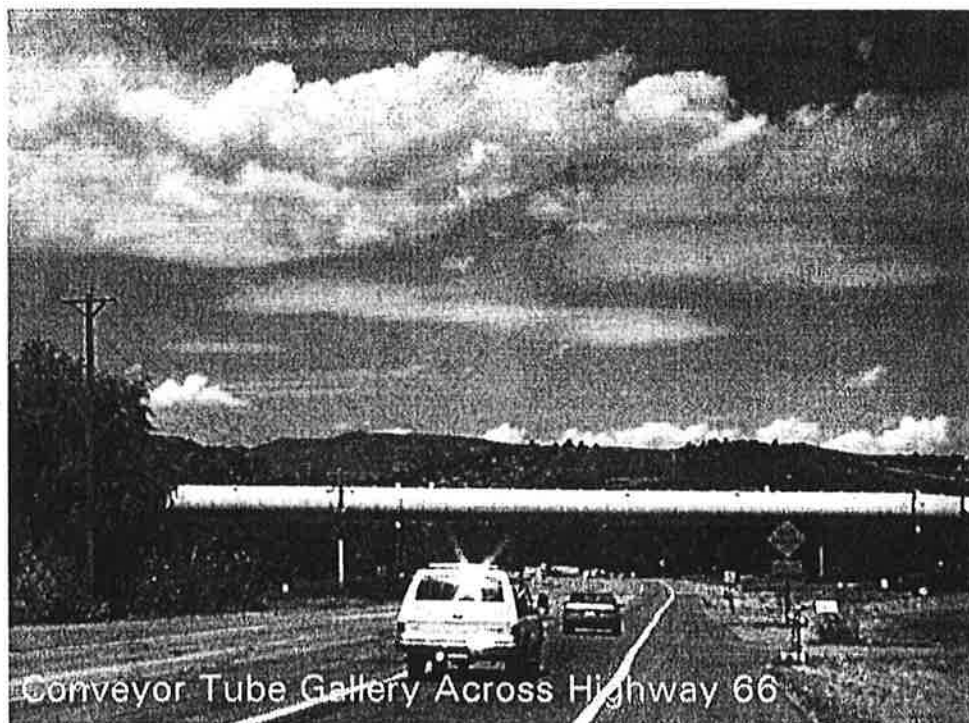
- No Vacation is Required
- County Transportation Needs Bridge for Public Access
- County Transportation and Southdown will Request an Amendment
- Southdown Uses Their New Bridge
- Public Access
 - ◆ Old Concrete Bridge with Reduced Limit
 - ◆ Emergency Access on Southdown Bridge

Donation Properties

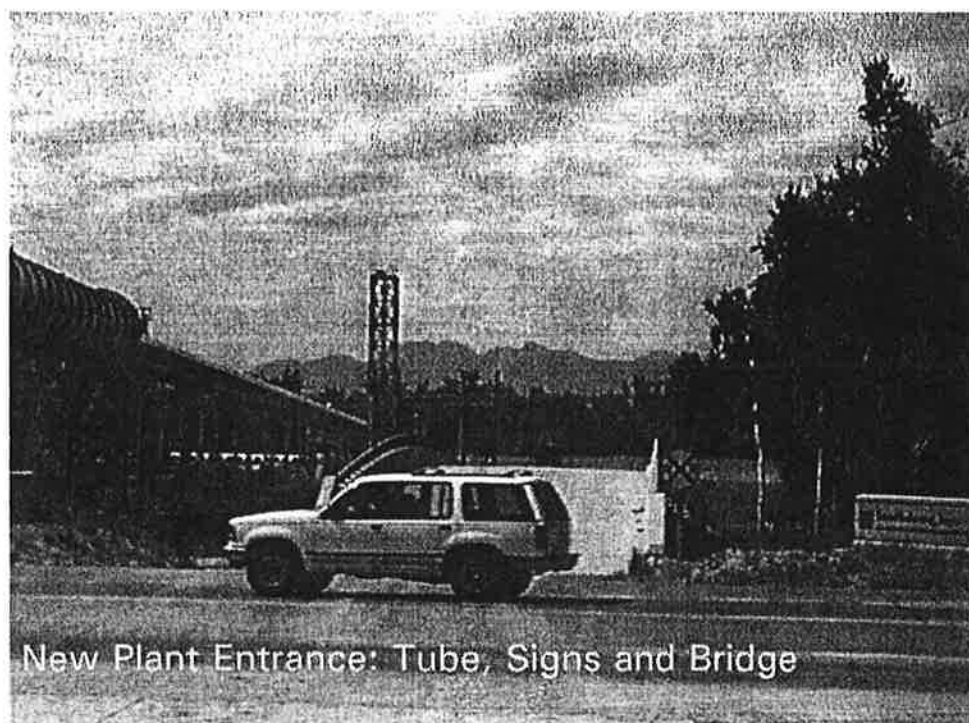
- 550 Acres Donated Before Sept 30, 1997
- Preconditions Satisfied
 - ◆ Commencement of Mining
 - ◆ Conveyor and Crusher in Service
 - ◆ County Acceptance of CR 47 & 49
- To be Named "Southdown Indian Mountain Open Space"

Sight Corridor-Hwy 66

- Paint Colors Approved by County
 - ◆ "Sea Horse" - Tube and Conveyor
 - ◆ "Green Gable" - Accents
- Landscape Approved by County
 - ◆ Signs Installed in 1996
 - ◆ Plantings Planned for October, 1997
 - ◆ Bid at \$60,000 - Native Plantings
 - ◆ Blend with Surroundings



Conveyor Tube Gallery Across Highway 66



New Plant Entrance: Tube, Signs and Bridge

Larimer County Quarry

- Last Loads Hauled: Jan, 1997
- 12,000 Highway Trips/year Eliminated
- 100 Trucks/Day on Haul Days
- 50 Acres Reclamation Complete

Southdown, Inc.

10/8/97

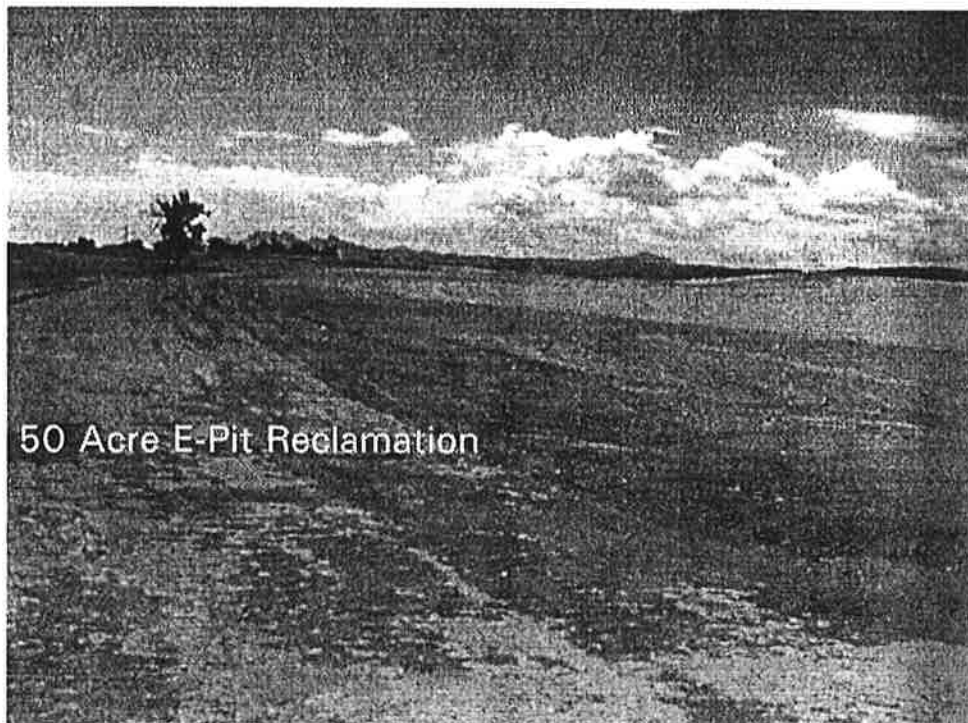
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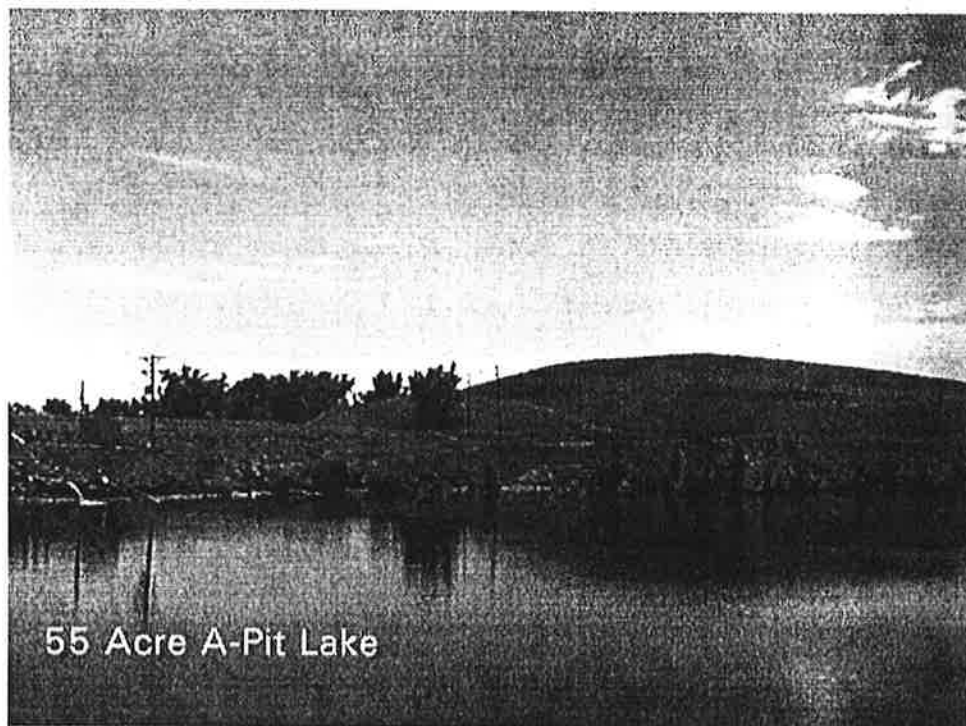
Larimer County Quarry Reclaimed

Plant Site Reclamation

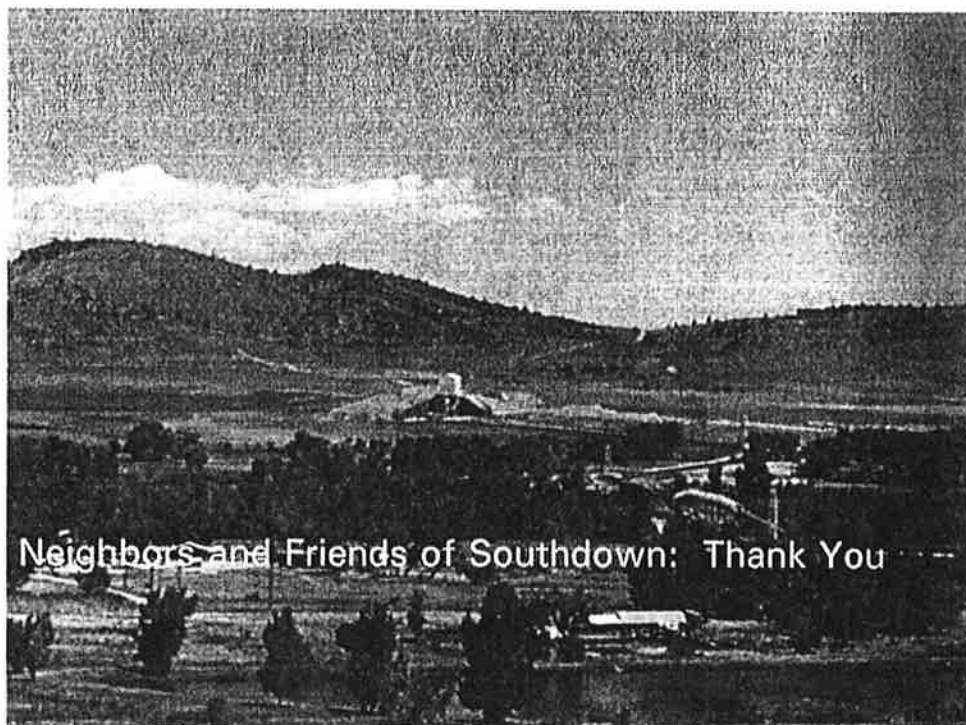
- 50 Acre E-Pit, South of Hygiene Road
 - ◆ Earth Moving Complete: 2.5 Million Yards
 - ◆ Planting in October, 1997
- 100 Acre A-Pit, Southeast Corner
 - ◆ 55 Acre Lake
 - ◆ 45 Acres of Grading next 18 Months



50 Acre E-Pit Reclamation



55 Acre A-Pit Lake



Neighbors and Friends of Southdown: Thank You

by Al Pace 10/14/91

Last Wednesday the Boulder County Commissioners renewed efforts by Southwestern Land Cement, a division of Southdown, Inc., to mitigate the effects of its mining operations on its neighbors and the environment. After a long process, the county allowed the cement plant to mine limestone and shale on Dow Flats north of Highway 66. County reviews are required for one, three and five years for every five years thereafter.

No one traveling into Lyons on Route 66 within the past year could fail to notice the enormous pipeline straddling the road. It closes a conveyor belt capable of delivering up to 800 tons of material per hour to the plant on the south side of the highway. The County approved four years corresponding to the company's need for limestone and shale types of shale. According to plant manager John Lohr, the shale, which lies on top of the limestone in strata inclined at an angle of 15 degrees, is rich in the silica and calcium necessary for the mixture of the cement.

To get at these layers, a D-11 bulldozer removes the overburden consisting of clay and poor-grade shale. This machine, the world's largest bulldozer, stands between 18 and 20 feet tall and has an 800 horse power engine.

During this phase, an archaeological monitor remains on site to observe the entire process and, in the event of uncovering artifacts, has the power to stop the work until a professional team excavates the entire find. Dow Flats lies between Indian Mountain and Rabbit Mountain, both of which have yielded significant Native American remains. Much of the topsoil is being used in the construction of the berms that demarcate the southern and eastern boundaries of the mine along Highway 66 and County Road 47 respectively. On weekdays visitors to Rabbit Mountain Open Space may observe some of Southdown's smaller bulldozers grading these embankments. Wooden posts set in concrete keep the mine from creeping beyond its boundaries to the north and west.

After the limestone or shale has been uncovered, drill rigs

make a hole five inches in diameter and 15 to 20 feet deep. Ammonium nitrate is used as an explosive since it can be mixed in the hole and the individual components are not, in themselves, volatile. Southdown uses a non-electric initiation of blast, which Lohr describes as "state-of-the-art." Thus car radios in the vicinity can't accidentally set off prematurely.

The D-11 bulldozer drives the excavated material into a pile for a front end loader with a bucket that holds up to 14 cubic yards, almost the world's largest. The loader fills Triple-7 Caterpillar dump trucks, each with a capacity of 100 tons, four times that of over-the-road trucks.

The material is dumped into the crusher house, an enormous sheet metal structure that sits above the crusher itself. This enclosure keeps the dust from escaping into the open air. "Our experience to date is that it's very effective," Lohr said.

The 1,000 horse power crusher engine propels horizontal impactor bars with hammer-like lugs. The impact causes the boulders of limestone and shale to virtually explode, reducing three-foot diameter chunks into four-inch pieces. Like the conveyor, it's attached to, it produces up to 800 tons per hour.

Next to the red shell of the crusher is an even larger blue receptacle, a dust collector that works like a giant vacuum machine. Southdown has installed dust collectors at every point along the conveyor where material

is transferred.

In order to satisfy the County's standards, the company has complied with a number of requirements, including: Dust collectors on all transfer points. Covers on the conveyor. A maximum of 120 acres exposed at any one time. Such areas to be reclaimed before opening another pit. Cease loading and hauling during sustained winds of 30 miles per hour. Wind speed indicators sound an alarm if winds attain that velocity for 15 minutes. Maximum drop height on the stacker to be 18 to 24 inches, and fogging sprays to reduce spread of dust.

Additionally, Southdown is currently modifying the dust collectors to make them even more effective.

The water rights attached to Dow Flats allow the company to maintain an extensive program for controlling dust on its roads. A water wagon with a 10,000 gallon capacity sprays and refills between eight and 10 times a day during dry weather days above 32 degrees Fahrenheit. Magnesium chloride is applied to all permanent roads once a year as is the practice with County maintained roads.

A noise control study conducted in July revealed a level well below the statutory limit of 55 dB, with a typical level of between 46 and 48 dB. According to Lohr, that's comparable to normal conversation. He also said that ambient noise from traffic and airplanes at that time registered above 55 dB.

Noise from the mining will also be muffled by berms on the southern and eastern boundaries of Dow Flats. As part of the agreement with the County, Southwestern had to rebuild County Road 47 since the road originally went through the mine. The berm along that corridor will be in place by the end of this month. "In two years, all mining will be out of the public's view from that road," Lohr said.

Areas not blocked by the berm will have been reclaimed by that time.

Native grasses are already established on the berm along the north side of Highway 66. Mature spruce trees have also been planted there. Other trees will follow, including crabapples, aspen, cottonwood and ponderosa pine.

Recently Southdown donated 550 acres in two parcels to County Open Space. The northern parcel will be open to the public in 2030 when reclamation of Dow Flats has been completed. A Denver based Native American group comprising several tribes holds cultural ceremonies on the southern parcel. It will remain closed to the public.

As an additional benefit to the public, the company has ceased its operations in Larimer County eliminating 100 truck trips per day. Fifty acres on the north side of Rabbit Mountain have been reclaimed. They used a cover crop of oats in this area to shade a mixture of pasture seedlings. The cover crop discourages sun-loving weeds from taking hold.

This Fall, Southwestern will plant a cover crop of rye on a fifty-acre reclamation of E pit on the south side of Highway 66. Other reclamation efforts in this